

DSA LAB 5

TASK1:

CODE:

```
#include<iostream>
#include<string>
using namespace std;
class Stack {
    int size=10;
    char arr[10];
    int top;
public:
    //constructor
    Stack():top(0){}
    //is empty function
    bool isEmpty()
    {
        return top == 0;
    }
    //is full fuunction
    bool isFull()
    {
        return top == 9;
    }
    //push value function
    void push(char x)
    {
        if (isFull())
        {
            cout << "Stack is full! bye bye";
            return;
        }
        arr[top] = x;
        top++;
    }
    //peek value function
    void peek(char &x)
    {
        if (isEmpty())
        {
            cout << "Stack is empty! bye bye";
            return;
        }
        x = arr[top];
    }
    //peek value function
    void pop()
    {
        if (isEmpty())
        {
```

```

        cout << "Stack is empty! bye bye";
        return;
    }
    top--;
}
void show()
{
    for (int i = 0; i < 10; i++)
    {
        cout << arr[i] << " ";
    }
}
};
int main()
{
    Stack stack;
    char ch;
    while (1)
    {
        system("cls");
        cout << "1. insert value\n2. pop value\n3. peek value\n4. is Empty\n5. is
Full\n6. show stack \n7. exit\n";
        int num;
        cin >> num;
        switch (num)
        {
            case 1:
                cout << "\nEnter character: ";
                cin >> ch;
                stack.push(ch);
                system("pause");
                break;
            case 2:
                stack.pop();
                system("pause");
                break;
            case 3:
                stack.peek(ch);
                cout << "character peek: " << ch;
                system("pause");
                break;
            case 4:
                stack.isEmpty();
                system("pause");
                break;
            case 5:
                stack.isFull();
                system("pause");
                break;
            case 6:
                stack.show();
                system("pause");
                break;
            case 7:
                exit(0);
                system("pause");
                break;
            default:

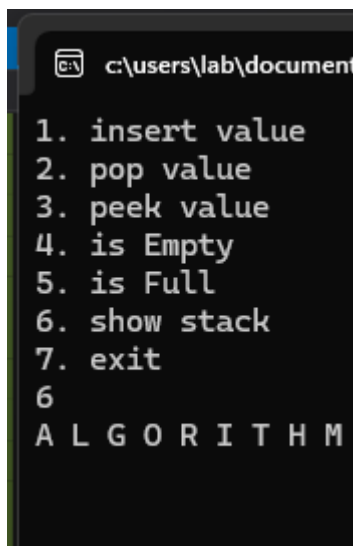
```

```

        break;
    }
}
}

```

OUTPUT:



```

c:\users\lab\document
1. insert value
2. pop value
3. peek value
4. is Empty
5. is Full
6. show stack
7. exit
6
ALGORITHM

```

TASK2:

CODE:

Part A

```

#include<iostream>
#include<string>
using namespace std;
class Stack {
    int size=10;
    int arr[10] = { 0 };
    int top;
public:
    //constructor
    Stack():top(0){}
    //is empty function
    bool isEmpty()
    {
        return top == 0;
    }
    //is full fuunction
    bool isFull()

```

```

{
    return top == 9;
}
//push value function
void push(int x)
{
    if (isFull())
    {
        cout << "Stack is full! bye bye";
        return;
    }
    arr[top] = x;
    top++;
}
//peek value function
void peek(int &x)
{
    if (isEmpty())
    {
        cout << "Stack is empty! bye bye";
        return;
    }
    x = arr[top];
}
//peek value function
void pop()
{
    if (isEmpty())
    {
        cout << "Stack is empty! bye bye";
        return;
    }
    top--;
}
//showing stack data
void show()
{
    for (int i = 0; i < 10; i++)
    {
        cout << arr[i] << " ";
    }
}
void reverse()
{
    if (isEmpty())
    {
        cout << "Stack is empty! bye bye";
        return;
    }
    int temp[10];
    for (int i = 0; i < 10; i++)
    {
        temp[i] = arr[i];
    }
    for (int i = 0; i < 10; i++)
    {
        pop();
    }
}

```

```

        top = 0;
        for (int i = 9; i >= 0; i--)
        {
            arr[top] = temp[i];
            top++;
        }
    };
int main()
{
    Stack stack;
    int n;
    while (1)
    {
        system("cls");
        cout << "1. insert value\n2. pop value\n3. peek value\n4. is Empty\n5. is
Full\n6. show stack \n7. reverse stack\n8. exit\n";
        int num;
        cin >> num;
        switch (num)
        {
            case 1:
                cout << "\nEnter number: ";
                cin >> n;
                stack.push(n);
                system("pause");
                break;
            case 2:
                stack.pop();
                system("pause");
                break;
            case 3:
                stack.peak(n);
                cout << "number peek: " << n;
                system("pause");
                break;
            case 4:
                stack.isEmpty();
                system("pause");
                break;
            case 5:
                stack.isFull();
                system("pause");
                break;
            case 6:
                stack.show();
                system("pause");
                break;
            case 7:
                stack.reverse();
                system("pause");
                break;
            case 8:
                exit(0);
                system("pause");
                break;
            default:
                break;
        }
    }
}

```

```

    }
}

```

Part b

```

#include<iostream>
#include<string>
using namespace std;
class Stack {
    int size=10;
    int arr[10] = { 0 };
    int top;
public:
    //constructor
    Stack():top(0){}
    //is empty function
    bool isEmpty()
    {
        return top == 0;
    }
    //is full fuunction
    bool isFull()
    {
        return top == 9;
    }
    //push value function
    void push(int x)
    {
        if (isFull())
        {
            cout << "Stack is full! bye bye";
            return;
        }
        arr[top] = x;
        top++;
    }
    //peek value function
    void peek(int &x)
    {
        if (isEmpty())
        {
            cout << "Stack is empty! bye bye";
            return;
        }
        x = arr[top];
    }
    //peek value function
    void pop()
    {
        if (isEmpty())
        {
            cout << "Stack is empty! bye bye";
            return;
        }
        top--;
    }
}

```

```

//showing stack data
void show()
{
    for (int i = 0; i < 10; i++)
    {
        cout << arr[i] << " ";
    }
}
void sort()
{
    if (isEmpty())
    {
        cout << "Stack is empty! bye bye";
        return;
    }
    int temp[10];
    for (int i = 0; i < 10; i++)
    {
        temp[i] = arr[i];
    }
    for (int i = 0; i < 10; i++)
    {
        pop();
    }
    top = 0;
    //sorting
    for (int i = 0; i < 9; i++)
    {
        for (int j = i+1; j < 9; j++)
        {
            if (temp[i] < temp[j])
            {
                int tem = temp[i];
                temp[i] = temp[j];
                temp[j] = tem;
            }
        }
    }
    for (int i = 9; i >= 0; i--)
    {
        arr[top] = temp[i];
        top++;
    }
}
};
int main()
{
    Stack stack;
    int n;
    while (1)
    {
        system("cls");
        cout << "1. insert value\n2. pop value\n3. peek value\n4. is Empty\n5. is
Full\n6. show stack \n7. sort stack\n8. exit\n";
        int num;
        cin >> num;
        switch (num)

```

```

{
case 1:
    cout << "\nEnter number: ";
    cin >> n;
    stack.push(n);
    system("pause");
    break;
case 2:
    stack.pop();
    system("pause");
    break;
case 3:
    stack.peek(n);
    cout << "number peek: " << n;
    system("pause");
    break;
case 4:
    stack.isEmpty();
    system("pause");
    break;
case 5:
    stack.isFull();
    system("pause");
    break;
case 6:
    stack.show();
    system("pause");
    break;
case 7:
    stack.sort();
    system("pause");
    break;
case 8:
    exit(0);
    system("pause");
    break;
default:
    break;
}
}

```

Part C

```

#include<iostream>
#include<string>
using namespace std;
class Stack {
    int size=10;
    int arr[10] = { 0 };
    int top;
public:
    //constructor
    Stack():top(0){}
    //is empty function

```



```

bool isEmpty()
{
    return top == 0;
}
//is full fuunction
bool isFull()
{
    return top == 9;
}
//push value function
void push(int x)
{
    if (isFull())
    {
        cout << "Stack is full! bye bye";
        return;
    }
    arr[top] = x;
    top++;
}
//peek value function
void peek(int &x)
{
    if (isEmpty())
    {
        cout << "Stack is empty! bye bye";
        return;
    }
    x = arr[top];
}
//peek value function
void pop()
{
    if (isEmpty())
    {
        cout << "Stack is empty! bye bye";
        return;
    }
    top--;
}
//showing stack data
void show()
{
    for (int i = 0; i < 10; i++)
    {
        cout << arr[i] << " ";
    }
}
//duplicate
void remove()
{
    if (isEmpty())
    {
        cout << "Stack is empty! bye bye";
        return;
    }
    int temp[10] = { 0 };
}

```

```

//removing duplicate
bool check[10] = { false };
int j = 0;
for (int i = 0; i < 9; i++)
{
    if (!check[arr[i]])
    {
        check[arr[i]] = true;
        temp[j] = arr[i];
        j++;
    }
}
for (int i = 0; i < 10; i++)
{
    pop();
}
top = 0;
for (int i = 0; i < 9; i++)
{
    arr[top] = temp[i];
    top++;
}
};
int main()
{
    Stack stack;
    int n;
    while (1)
    {
        system("cls");
        cout << "1. insert value\n2. pop value\n3. peek value\n4. is Empty\n5. is
Full\n6. show stack \n7. remove duplicate\n8. exit\n";
        int num;
        cin >> num;
        switch (num)
        {
            case 1:
                cout << "\nEnter number: ";
                cin >> n;
                stack.push(n);
                system("pause");
                break;
            case 2:
                stack.pop();
                system("pause");
                break;
            case 3:
                stack.peek(n);
                cout << "number peek: " << n;
                system("pause");
                break;
            case 4:
                stack.isEmpty();
                system("pause");
                break;

```

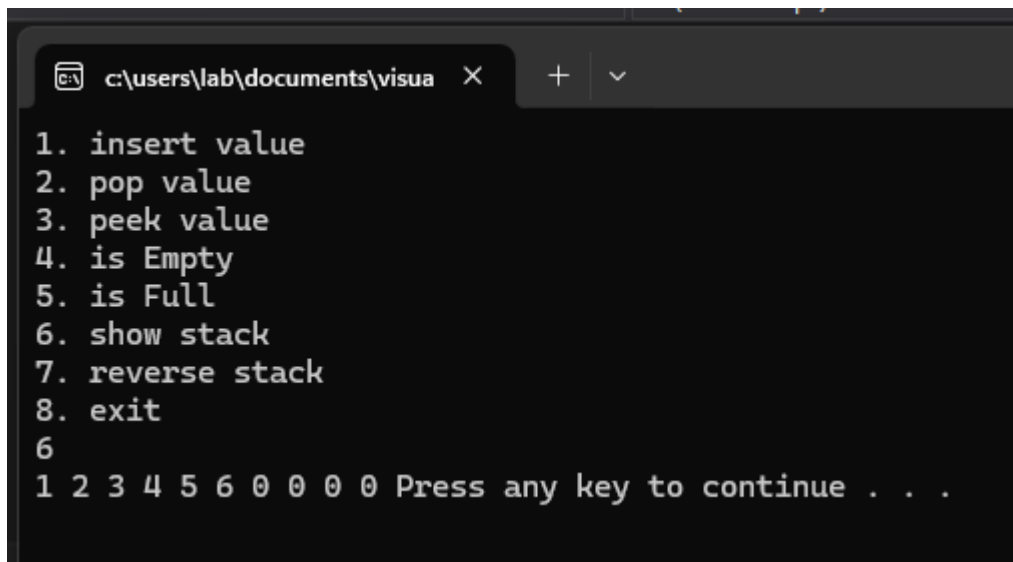
```

        case 5:
            stack.isFull();
            system("pause");
            break;
        case 6:
            stack.show();
            system("pause");
            break;
        case 7:
            stack.remove();
            system("pause");
            break;
        case 8:
            exit(0);
            system("pause");
            break;
        default:
            break;
    }
}
}

```

OUTPUT:

Part A



```

c:\users\lab\documents\visua X + v
1. insert value
2. pop value
3. peek value
4. is Empty
5. is Full
6. show stack
7. reverse stack
8. exit
6
1 2 3 4 5 6 0 0 0 0 Press any key to continue . . .

```

```
c:\users\lab\documents\visua X + v
1. insert value
2. pop value
3. peek value
4. is Empty
5. is Full
6. show stack
7. reverse stack
8. exit
6
0 0 0 0 6 5 4 3 2 1 Press any key to continue . . .
```

Part b

```
c:\users\lab\documents\visua X + v
1. insert value
2. pop value
3. peek value
4. is Empty
5. is Full
6. show stack
7. sort stack
8. exit
6
5 4 8 1 3 9 0 0 0 0 Press any key to continue . . . |
```

```
c:\users\lab\documents\visua X + v
1. insert value
2. pop value
3. peek value
4. is Empty
5. is Full
6. show stack
7. sort stack
8. exit
6
0 0 0 0 1 3 4 5 8 9 Press any key to continue . . . |
```

Part C

```
c:\users\lab\documents\visua X + v
1. insert value
2. pop value
3. peek value
4. is Empty
5. is Full
6. show stack
7. remove duplicate
8. exit
6
2 2 3 4 5 6 6 4 7 0 Press any key to continue . . .
```

```
c:\users\lab\documents\visua X + v
1. insert value
2. pop value
3. peek value
4. is Empty
5. is Full
6. show stack
7. remove duplicate
8. exit
6
2 3 4 5 6 7 0 0 0 0 Press any key to continue . . . |
```

TASK3:

CODE:

```
#include<iostream>
#include<string>
using namespace std;
struct Node {
    float cgpa;
    int roll;
    int marks;
    Node* next;
    Node():next(nullptr){}
};
class Linkelist {
    Node* head = nullptr;
    Node* newnode = nullptr;
    Node* current = nullptr;
    int count = 0;
public:
    Linkelist() {}
    //destructor
    ~Linkelist()
    {
        current = head;
        newnode = nullptr;
        while (current != nullptr)
        {
            newnode = current->next;
            delete current;
            current = newnode;
        }
        current = nullptr;
        newnode = nullptr;
        delete current;
        delete newnode;
    }
    void add()
    {
```

```

newnode = new Node();
int roll;
cout << "enter roll no: ";
cin >> roll;
current = head;
while (current != nullptr)
{
    if (current->roll == roll)
    {
        cout << "data cannot be entered roll no is same!";
        return;
    }
    current = current->next;
}
newnode->roll = roll;
cout << "enter cgpa: ";
cin >> newnode->cgpa;
cout << "\nenter marks one by one of all subjects: ";
int num, sum = 0;
for (int i = 0; i < 6; i++)
{
    cin >> num;
    sum = sum + num;
}
newnode->marks = sum;
if (head == nullptr)
{
    head = newnode;
}
else
{
    current = head;
    head = newnode;
    head->next = current;
}
}
//print
void print()
{
    current = head;
    while (current != nullptr)
    {
        cout << "\n\nRoll No: " << current->roll << "\nCGPA: " << current-
>cgpa << "\nMarks: " << current->marks;
        current = current->next;
    }
}
};
class Stack {
    Linklist list[10];
    int size=10;
    int top;
public:
    //constructor
    Stack():top(0){}
    //is empty function
    bool isEmpty()
    {

```

```

        return top == 0;
    }
    //is full fuunction
    bool isFull()
    {
        return top == 9;
    }
    //push value function
    void push()
    {
        if (isFull())
        {
            cout << "Stack is full! bye bye";
            return;
        }
        list[top].add();
        top++;
    }

    //peek value function
    void pop()
    {
        if (isEmpty())
        {
            cout << "Stack is empty! bye bye";
            return;
        }
        top--;
    }
    //showing stack data
    void show()
    {
        for (int i = 0; i < 10; i++)
        {
            list[i].print();
            cout << "\n\n";
        }
    }
};

int main()
{
    Stack stack;
    int n;
    while (1)
    {
        system("cls");
        cout << "1. insert node\n2. pop node\n3. is Empty\n4. show stack \n5.
exit\n";
        int num;
        cin >> num;
        switch (num)
        {
            case 1:
                stack.push();
                system("pause");
                break;
            case 2:

```



```
        stack.pop();
        system("pause");
        break;

    case 3:
        stack.isEmpty();
        system("pause");
        break;
    case 4:
        stack.show();
        system("pause");
        break;

    case 5:
        exit(0);
        system("pause");
        break;
    default:
        break;
    }
}
```

OUTPUT:

```
c:\users\lab\documents\visua X + v
1. insert node
2. pop noden3. is Empty
4. show stack
5. exit
4

Roll No: 3035
CGPA: 4
Marks: 600

Roll No: 1606
CGPA: 4
Marks: 540
```

TASK4:

CODE:

```
#include<iostream>
#include<string>
using namespace std;
class Stack {
    int size = 10;
    string arr[10] = { " " };
    int top;
public:
    //constructor
    Stack() :top(0) {}
    //is empty function
    bool isEmpty()
    {
        return top == 0;
    }
    //is full fuunction
    bool isFull()
    {
        return top == 9;
    }
}
```

```

}
//push value function
void push(string x)
{
    if (isFull())
    {
        cout << "Stack is full! bye bye";
        return;
    }
    arr[top] = x;
    top++;
}
//peek value function
void current()
{
    if (isEmpty())
    {
        cout << "Stack is empty! bye bye";
        return;
    }
    cout << "current element is: ";
    cout << arr[top];
}
//peek value function
void pop()
{
    if (isEmpty())
    {
        cout << "Stack is empty! bye bye";
        return;
    }
    top--;
}
//showing stack data
void show()
{
    for (int i = 0; i < 10; i++)
    {
        cout << arr[i] << " ";
        cout << endl;
    }
}
//clear history
void clear()
{
    for (int i = 0; i < 10; i++)
    {
        pop();
    }

    top = 0;
}
//forward navigate
void forward()
{
    if (arr[top + 1] == " ")
    {
        cout << " \nforward element is empty!";
    }
}

```

```

    }
    else
    {
        top++;
        cout << "\nforward elemnt: ";
        cout << arr[top];
    }
}
//backward navigate
void backward()
{
    top--;
    cout << "\nbackward elemnt: ";
    cout << arr[top];
}
};
int main()
{
    Stack stack;
    int n;
    string num;
    while (1)
    {
        system("cls");
        cout << "1. insert URL\n2. pop URL\n3. forward navigate\n4. backward\n5. current elemnt\n6. show stack\n7. clear history \n8. exit\n";
        cin >> n;
        switch (n)
        {
            case 1:
                cout << "\nEnter URL: ";
                cin.ignore();
                stack.push(num);
                getline(cin, num);
                system("pause");
                break;
            case 2:
                stack.pop();
                system("pause");
                break;
            case 3:
                stack.forward();
                system("pause");
                break;
            case 4:
                stack.backward();
                system("pause");
                break;
            case 5:
                stack.current();
                system("pause");
                break;
            case 6:
                stack.show();
                system("pause");
                break;
            case 7:

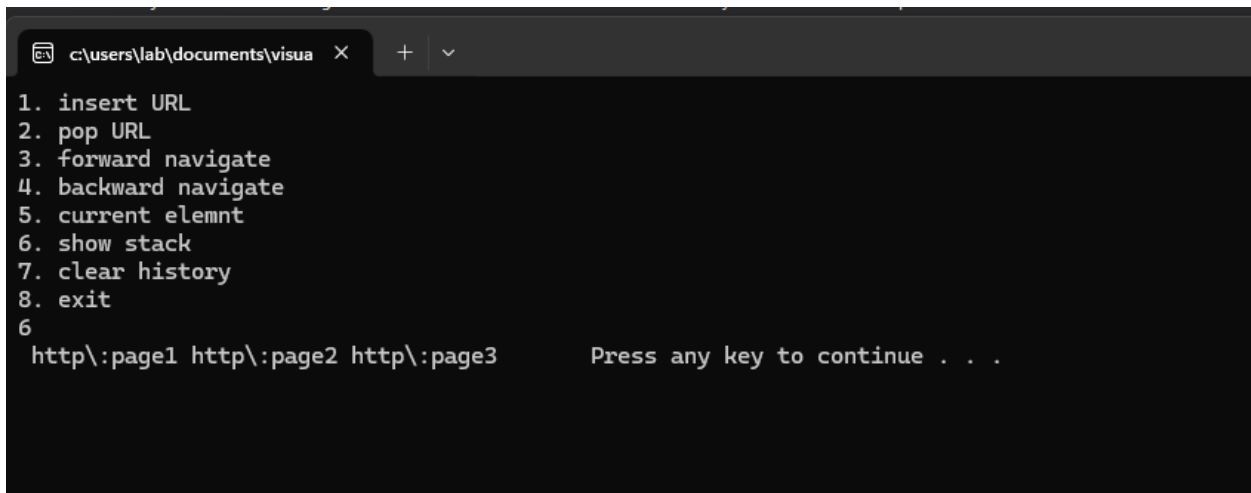
```

```

        stack.clear();
        system("pause");
        break;
    case 8:
        exit(0);
        system("pause");
        break;
    default:
        break;
    }
}
}

```

OUTPUT:



```

c:\users\lab\documents\visua X + v
1. insert URL
2. pop URL
3. forward navigate
4. backward navigate
5. current elemnt
6. show stack
7. clear history
8. exit
6
http\:page1 http\:page2 http\:page3      Press any key to continue . . .

```